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ABSTRACT

Time-modulated quantum electrodynamics (TM-QED) investigates the dynamic interactions between quantum emitters (QEs) and photonic systems with temporally varying properties, allowing for unprecedented control over light–matter interactions. This paper presents a framework for employing TM-QED in dynamic systems by coupling the Gorini–Kossakowski–Sudarshan–Lindblad equation with the time-Floquet method and Mie theory. The proposed approach enables the study of QEs interacting with dynamically modulated photonic structures, focusing on the dynamic control of the density of states and coupling between frequency harmonics. Unlike traditional static photonic environments, our approach enables active manipulation of emission properties post-fabrication, offering new avenues for quantum control. By leveraging time modulation, we demonstrate the generation of frequency harmonics, enhanced spontaneous emission rates, and control over the radiation of quantum emitters. We also demonstrate how plasmonic resonances enhance light–matter interactions, resulting in optimized spectral properties. Time modulation offers a fourth dimension in optimizing quantum structures, which can be utilized to manipulate emission properties post-fabrication actively. Finally, understanding the dynamics of harmonic coupling in QE's states can help improve quantum sensing, single-photon sources, and read-out.

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I. INTRODUCTION

In the past few years, intensive research has been dedicated to studying new approaches to efficiently control the emission of quantum emitters (QEs) with applications as diverse as lasers, LEDs, quantum sensors, quantum communication systems;[1,2] single-photon-sources;[3] free-space optical communications; and others.[4,5] The most efficient way of controlling the emission of QEs (such as quantum dots, quantum wells, and fluorescence molecules) is by engineering their photonic density of states (PDoS) since it affects the QE emission properties, namely, spontaneous emission rate (Γ), radiated power (Q), quantum efficiency (η), and radiation pattern (D) or scattering (S). According to the Fermi golden rule, enhancing the PDoS increases the probability for the QE to decay, allowing a faster decay rate Γ. The ratio between Γ and the emission rate in free space (Γ_0) is known as the Purcell factor (P).[6,7] Thus,

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manipulating P is crucial for several applications since it implies controlling the QE lifetime and modulation speed. The most common approach for managing its properties is placing the QE close to or inside resonant photonic structures, such as photonic crystal cavities,[8] dielectric nanoholes,[9] nano-antennas, or hyperbolic metamaterials (HMM).[5,10] In addition, spatially periodic media, such as photonic crystals and metasurfaces, are also effective in engineering the local density of electromagnetic states and controlling the spontaneous emission of QEs.[11–13] By analogy with spatial periodicity, temporal modulation can be introduced into photonic systems, adding a new degree of freedom to the design paradigm and enabling control over both the spatial and spectral characteristics of QE radiation. Nonetheless, after the fabrication, the system is static, and radiation manipulation can only be achieved by changing QE's excitation parameters.

Time-modulated systems have emerged as a promising area in quantum electrodynamics, particularly in applications involving the manipulation of coherence and thermal radiation at the nanoscale.[14–18] When integrated with photonic structures, these systems enable spatial and temporal control over radiated wave properties.[19,20] For example, in near-field regimes, time modulation can influence thermal radiation coherence and allow dynamic control of emission rates, offering new avenues for controlling light–matter interactions and spontaneous emission processes.[21–25] In Ref. 23, the authors used frequency-modulated optical transitions for spectral engineering of single-photon emission. They demonstrate that a two-level system modulated faster than its optical lifetime can produce single photons with a tunable spectrum. Moreover, it has also been shown that the spatial coherence of thermal radiation can be controlled using time-modulated photonic systems that support surface polaritons (SPPs).[25–28] Micro- and nano-particles composed of plasmonic materials can support geometrical, or Mie type, resonances that not only enhance light–matter interactions for a more profound impact of time modulation but also enable control over key features such as directionality and orbital or spin angular momentum.[29–31] A time-modulated structure allows for controllable energy transfer between different frequencies (change of the photon momentum), enabling new correlations between frequency components.

Open quantum systems can be described using one of the general forms of the Markovian master equation,[31–35] the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) equation to model the dynamics of the quantum states, where spontaneous emission is considered via a Lindblad superoperator.[32] Quantum field theoretical methods have been employed to study light–matter interactions in complex photonic systems, such as the quantum-optical description of emitter–plasmon coupling and polaritonic modes in metallic nanostructures,[33] electrochemically switchable multimode strong coupling in plasmonic nanocavities,[34] and the control of strong light–matter coupling in hybrid plasmonic–dielectric environments.[35] The interaction between the QE and the electric field can be modeled under dipolar approximation, which can be coupled to time-varying fields. In a multi-level system, the QE interacts with a time-modulated electric field, which introduces multiple harmonics into the system, leading to intricate coherence effects and harmonic generation. Unlike previous studies that model the quantum emitter as a classical dipole, where the emitter is treated as a point source with a fixed polarizability or dipole moment,[36–38] our approach provides a quantum-mechanical treatment by directly solving the general form of the Markovian master equation. In classical dipole models, the quantum dynamics of the emitter, such as population transfer and coherence between states, are not explicitly captured and the feedback between the emitter and its environment is treated in a simplified. Some studies have coupled GKSL-like quantum dynamics with static electromagnetic solvers to capture the quantum-classical interaction better.[39–41] However, our study integrates a time-modulated Floquet–Mie framework with GKSL formalism, enabling the modeling of quantum emitters embedded in dynamically varying photonic structures. This enables us to investigate how the time modulation of the environment directly affects quantum coherence, sideband generation, and frequency-selective emission, capabilities that surpass those achievable by traditional static or classical models. In the case of a three-level system, the interaction is also notable as it involves transitions between multiple energy states with distinct frequencies.

In this work, we present a comprehensive framework for modeling and analyzing time-modulated quantum electrodynamics (TM-QED) in dynamic systems by coupling the GKSL equations with an adequate electromagnetic formalism. Several methods exist for coupling quantum systems with electromagnetic fields.[39–41] In Refs. 28 and 29, the authors developed a finite-difference time domain (FDTD)-style algorithm that uses a photon wave-function in the spin-1 Dirac formulation to directly discretize space and time, tracking the spatio-temporal field evolution at the level of single-photon states coupled to the quantum dot in a fully quantized description, which allows for tracking of the Heisenberg uncertainty principle during the spontaneous emission of a single photon. The time-floquet Mie theory is not a direct time-domain solver such as FDTD, i.e., it does not perform time stepping. Instead, it describes the coupling of a quantum emitter to the classical electromagnetic field using Mie theory,[42,43] enabling the calculation of dipolar emission, the Purcell factor, and quantum coherence in systems where quantum emitters interact with dynamically modulated plasmonic structures. While Tafur and Leuenberger's FDTD-based method can also describe time-modulated electromagnetic systems, it does not incorporate decoherence or dissipative mechanisms and becomes computationally expensive for low modulation speeds. Meanwhile, the quantized photon's spontaneous emission is defined as a coherent process in Refs. 28 and 29; here, we describe the spontaneous (radiative) decay of the quantum emitter employing the incoherent transition rate in the Lindblad-type equations. Previous studies have shown that periodically driven open quantum systems governed by Markovian dynamics can be described within the Lindblad formalism using a Floquet-based approach, where the Lindblad master equation admits a time-periodic solution due to the Floquet structure of the Liouvillian superoperator[44,45] In this sense, the proposed approach enables the calculation of dipolar emission, Purcell factor, and quantum coherence in systems where quantum emitters interact with dynamically modulated photonic structures. We demonstrate how time modulation can couple harmonic frequencies, enhance spontaneous emission rate, and provide real-time control over emission dynamics. Furthermore, as an illustrative example, we explore the role of resonant plasmonic effects in amplifying light–matter interactions and propose optimization strategies for frequency-selective emission using the proposed

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formalism. Our approach enables optimizing time-modulated systems by incorporating the coupling between quantum emitters (QEs) and their environment. This allows for targeted enhancements, such as increasing the Purcell factor and improving overall emission efficiency, as in Ref. 28, while facilitating control over the interaction dynamics within complex quantum photonic structures. Ranging from classical to quantum mechanical descriptions,[46,47] our paper offers a comprehensive understanding together with a theoretical framework for studying the interaction of an open quantum system with temporally modulated electromagnetic fields.

II. TIME-MODULATED QUANTUM EMITTER

For the proposed modeling, we consider an N-level ($|n\rangle$ where the ith state has energy $\hbar\omega_i$. To model the quantum states of the emitter, we resort to the open form of one of the general form of the Markovian master equation, which is the Lindbladian or GKSL equation in the Lindblad form,[44,45,48–50]

$$\frac{d\hat{\rho}(t)}{dt} = \frac{i}{\hbar}\left[\hat{\rho}(t), \hat{H}\right] + \mathcal{L}_\gamma \hat{\rho}(t), \tag{1}$$

where $\hat{\rho}$ is the density operator, $\hat{H}$ is the Hamiltonian, and $\mathcal{L}_\gamma$ denotes a superoperator accounting for dissipative processes such as spontaneous emission, under the assumption of negligible memory effects (i.e., Markovian evolution). Under the dipolar approximation, the semi-classical light–atom interaction Hamiltonian and $\hat{H}$ can be written, respectively, as

$$\begin{aligned}\hat{H} &= \hat{H}_{atom} + \hat{H}_{atom:field}(t) \\ &= \sum_i \hbar\omega_i |i\rangle\langle i| + \sum_{i\neq j} |i\rangle \frac{\hbar}{2}\hat{d}_{ij}\cdot \mathbf{E}(\vec{r},t)\langle j| e^{j\omega_{ij}t},\end{aligned} \tag{2}$$

$$\mathcal{L}_\gamma\hat{\rho}(t) = \sum_{i,j}\gamma_{ij}\left(\left[\hat{\sigma}_{ij}, \hat{\rho}\hat{\sigma}_{ij}^\dagger\right] + \left[\hat{\sigma}_{ij}\hat{\rho}, \hat{\sigma}_{ij}^\dagger\right]\right), \tag{3}$$

where $\omega_{ij} = \omega_i - \omega_j$, $\hat{d}_{ij}$ is the dipolar operator, $\hat{H}_{atom}$ takes into account the atom/system response, $\hat{H}_{atom:field}$ is the dipolar approximation of the atom/field interaction, $\mathbf{E}(\mathbf{r},t)$ is the electric field, γ_{ij} is the dumping frequency between ith and jth levels, and $\hat{\sigma}_{ij}$ is the ladder operator. Considering a time-modulated electric field, $\mathbf{E}(\mathbf{r},t)$ can be written as

$$\mathbf{E}(\mathbf{r},t) = \sum_{i\neq j}\left(\sum_n E_{ij}^n e^{j\left(\vec{k}_{ij}^n\vec{r} - \omega_{ij}^n t\right)}\boldsymbol{\varepsilon}_{\mathbf{ij}}^{\mathbf{n}}\right), \tag{4}$$

where i and j are related to the transition between the states, $\omega_{ij}^n = \omega_{ij} + \Delta_{ij} + n\omega_m$ is the frequency related to the ω_{ij} transition, ω_m is the modulation frequency, Δ_{ij} stands for the frequency shift, E_{ij}^n is the nth harmonic electric field amplitude, and $\boldsymbol{\varepsilon}_{\mathbf{ij}}^{\mathbf{n}}$ is the versor of the electric field. To solve the linear system, we can resort to the Floquet-eigen solutions of (1),

$$\rho(t) = \sum_n\sum_m A_m B_{n,m}\rho^{m,n} e^{\lambda_m t + jn\omega_m t}, \tag{5}$$

where λ_m represents the complex quasienergies (Floquet–Liouville exponents), $B_{n,m}$ is the associated element of the eigenvector, and A_m is the eigenvector amplitude. In this sense, by truncating the number of time-harmonics from $-M$ to M, Eq. (1) can be rewritten as a matrix,

$$\frac{d\rho_M(t)}{dt} = \mathbf{M}\rho_M(t), \tag{6}$$

where $\rho_M(t) = \left[\rho_M^{-M}; \rho_M^{-M+1}; \ldots; \rho_M^0; \ldots; \rho_M^M\right]$, and ρ_M^n corresponds to the nth time-harmonic and is equal to $\rho_M^n = \left[\rho_{11}^n; \rho_{22}^n; \ldots; \rho_{NN}^n; \rho_{12}^n; \ldots; \rho_{1N}^n; \rho_{21}^n; \ldots \rho_{N,N-1}^n\right]$, with $\rho_{i,j}^n$ being the nth harmonic of $\hat{\rho}_{ij}$. $\mathbf{M}$ is defined as an $N^2(2M+1)$ square matrix obtained by resorting to (1)–(3). We provide the full formalism for two and three level systems in the supplementary material. After calculating the density of states, we can exploit the dipolar approximation to calculate the dipolar transition moment from level i to j, as follows:[51,52]

$$\begin{aligned}\mathbf{p}_{\mathbf{ij}}(t) &= \mu_{ij}\rho_{ij}\boldsymbol{\varepsilon} e^{-j\omega_{ij}t} \\ &= N_{emi}\mu_{ij}\left(\sum_n\sum_m A_m B_{n,m}\rho_{ij}^{m,n} e^{(\lambda_m + j(\omega + n\omega_m))t}\right)\boldsymbol{\varepsilon}_{\mathbf{ij}}^{\mathbf{n}},\end{aligned} \tag{7}$$

in which N_{emi} is the number of quantum emitters. Moreover, in (7), the term with frequency $\omega_{ij} + \omega + n\omega_m$ is not considered due to the high frequency. Another important consideration is that as $t \to \infty$, modes with real parts vanish, leaving only purely imaginary eigenvalues. Therefore, we only need to consider purely imaginary eigenvalues in the steady-state analysis. Since $\hat{\rho}(t)$ is written by its eigenvalues Eq. (5), the condition $\frac{d\hat{\rho}(t)}{dt} \to 0$ means that each state population harmonic converges to a different steady state and the emission and population of each state remain time-modulated although in a steady state. Having calculated the dipolar moments, we can couple the QE to the electrodynamic system, as presented in Fig. 1. In the proposed system, the electric field at the QE position is the composition of the incident electric field $[\mathbf{E_i}(\mathbf{r},t)]$ with the electric field generated by the interaction with the nearby structures $[\mathbf{E_r}(\mathbf{r},t)]$, considering the QE itself also radiates and generates electric field $[\mathbf{E_{QE}}(\mathbf{r},t)]$, as can be seen in Fig. 1(a). In this sense,

$$\vec{\mathbf{E}}(\mathbf{r},t) = \mathbf{E_i}(\mathbf{r},t) + \mathbf{E_{QE}}(\mathbf{r},t) + \mathbf{E_r}(\mathbf{r},t). \tag{8}$$

Using dipolar approximation, the radiation of the QE can be written as a dipolar emission, where the dipole moment is given in Eqs. (7) and (8). In (9), $\mathbf{E_r}(\mathbf{r},t) = \mathbf{E_r^i}(\mathbf{r},t) + \mathbf{E_r^{QE}}(\mathbf{r},t)$ is a contribution of the structure interacting with the incident light $\mathbf{E_r^i}(\mathbf{r},t)$ and the $\mathbf{E_r^{QE}}(\mathbf{r},t)$. Both can be calculated using conventional time-modulated solvers, such as time-modulated rigorous coupled wave (RCWA) analysis;[53] time-modulated discrete dipole approximation (DDA);[54] and the time-Floquet Mie theory[42,55] and time varying crystals theory.[56] It is worth noting that the total $\mathbf{E}(\mathbf{r},t)$ depends on the states to calculate the dipolar approximation, while the states need the Electric field, which can be seen in the QE energy levels shown in Fig. 1(b). In this sense, we make use of the Newton–Raphson[57] algorithms to converge the density of states, which provides the exact same electric field necessary to produce $\mathbf{p}_{\mathbf{ij}}(t)$. After the calculations, we can compute the scattering properties of the time-modulated QE, as the example shown in Fig. 1(c). A chart of the procedure can be seen in Fig. 2, where we start by calculating the background electric fields $\mathbf{E_i}(\mathbf{r},t)$ and $\mathbf{E_r^i}(\mathbf{r},t)$, and computing the density of states ρ along with the dipole moments $\mathbf{p}_{\mathbf{ij}}$.

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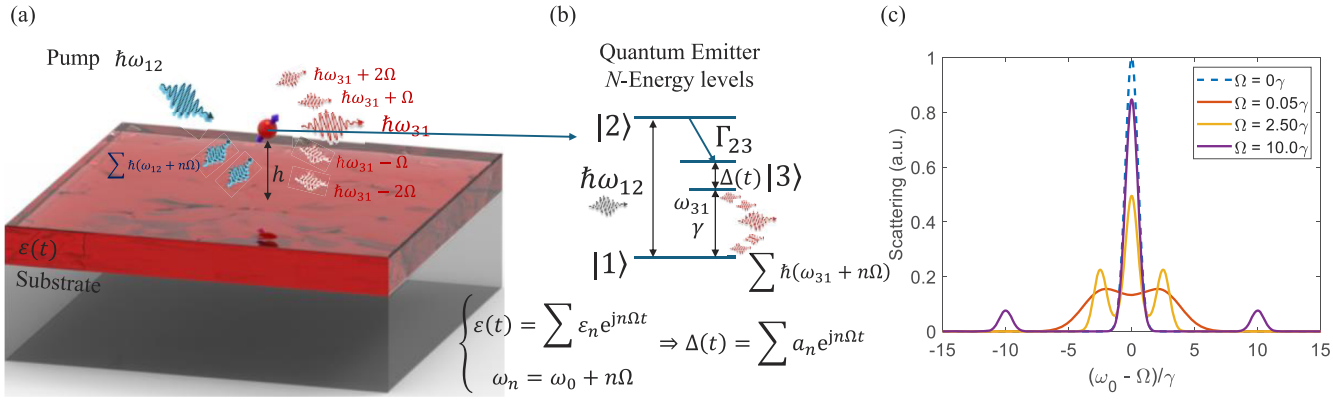


FIG. 1. (a) Schematic of a quantum emitter (QE) interacting with a time-modulated structure (the red slab above the substrate), where the total electric field $\mathbf{E}(\mathbf{r},t)$ at the QE is a sum of the incident field $\mathbf{E_i}(\mathbf{r},t)+\mathbf{E_r^i}(\mathbf{r},t)$, the field radiated by the $\mathbf{E_{QE}}(\mathbf{r},t)+\mathbf{E_r^{QE}}(\mathbf{r},t)$, and the field reflected by the structure. (b) Energy-level diagram of the QE, showing transitions influenced by the modulated electric field, crucial for achieving steady-state solutions. (c) Scattering spectrum as a function of frequency detuning, illustrating the impact of modulation strength Ω on the emission profile.

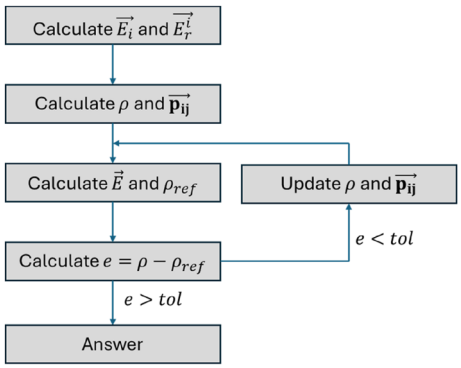


FIG. 2. Flow chart illustrating the iterative procedure to achieve the steady-state solution of the time-modulated QE system.

With this information, we capture the total electric field radiated by the QE and reflected by the structure [as shown in Fig. 1(a)] and calculate the new ρ_{ref}. In sequence, with the error $e=\rho-\rho_{ref}$ (note that all states levels need to converge) being smaller than a defined tolerance *tol*, the present state is correct; otherwise, we use the Newton method to calculate the new ρ and perform another interaction until the approach converges. It is also important to highlight that, in our model the coupling between harmonics originates from the electric fields that excite the time-harmonic components in the QE state. In this context, the energy exchange between harmonics is governed by the properties of the time-modulated material, which changes the photon momentum and, consequently, their frequencies. Hence, the QE is excited by different electric field harmonics Eq. (4), and each harmonic operates as a detuned excitation for the QE.

III. POLARIZABILITY ANALYSIS

To validate our approach, we calculate the polarizability of a two- and three-level quantum system, considering a low-power regime. Under these considerations, the nth harmonic of the polarizability of a two-level system can be calculated as

$$\alpha_n(\omega)=\frac{2d_{12}^2}{\hbar}\frac{1}{\left[(\omega-n\omega_m)^2-\omega_{12}^2+\mathrm{j}\Gamma_{12}(\omega-n\omega_m)\right]}, \tag{9}$$

where ω is the excitation light angular frequency. By considering $\omega_0=\omega_{12}$, $\Gamma_{12}=\Gamma/20$, and $\omega_m=\omega_0/5$, the real (solid lines) and imaginary (dashed lines) α_n for a two-level system can be seen in Fig. 3(a) for $n=(-2,2)$ [energy levels shown as an inset in Fig. 3(b)]. As illustrated, polarizability is related to a Lorentzian-shaped curve, as expected and demonstrated in other studies,[58] which is verified in our formalism. Moreover, we also calculate the ratio between coherent and spontaneous emission, given by $\rho_{12}\rho_{12}^*/\rho_{22}$[51] when all harmonics of the excitation have the same amplitude, as shown in Fig. 3(b). As can be seen, time modulation decreases the coherence of a two-level system since some photons decay out of coherence in different harmonics.

For a V-shaped three-level system polarizability, we considered $\omega_{13}=2\omega_{12}$, $\Gamma_{12}=\Gamma_{13}=\Gamma_{23}/20=\omega_{13}/100$, $\omega_m=\omega_{13}/5$, and finally, the Rabi frequency $\Omega_{12}=2\Omega_{13}$. Ω_{12} excites the electrons to the 2-level, which decays non-radiatively to 3-level. Finally, the radiative decay from level 2 to 1 is stimulated by an external time-modulated electric field, with equal harmonic amplitudes. As a consequence, each frequency harmonic can be associated by a polarizability α_n that relates the excitation with the emission of the QE. The real (solid lines) and imaginary (dashed lines) α_n are given in Fig. 3(c) for $n=(-2,2)$ [energy levels shown as an inset in Fig. 3(d)], and as expected, the polarizability has a Lorentzian shape for all harmonics. When subjected to a time-modulated electric field, the quantum emitter will emit at frequencies corresponding to all harmonics, with varying amplitudes depending on the detuning. For a three-level system, it is considerably more challenging to obtain coherence since it needs to invert the population on E_3. However, introducing time-modulation helps with coherence, as shown in Fig. 3(d). It is important to note that when a time-modulated electric field excites the QEs with different time harmonics, the QEs behave as multiple

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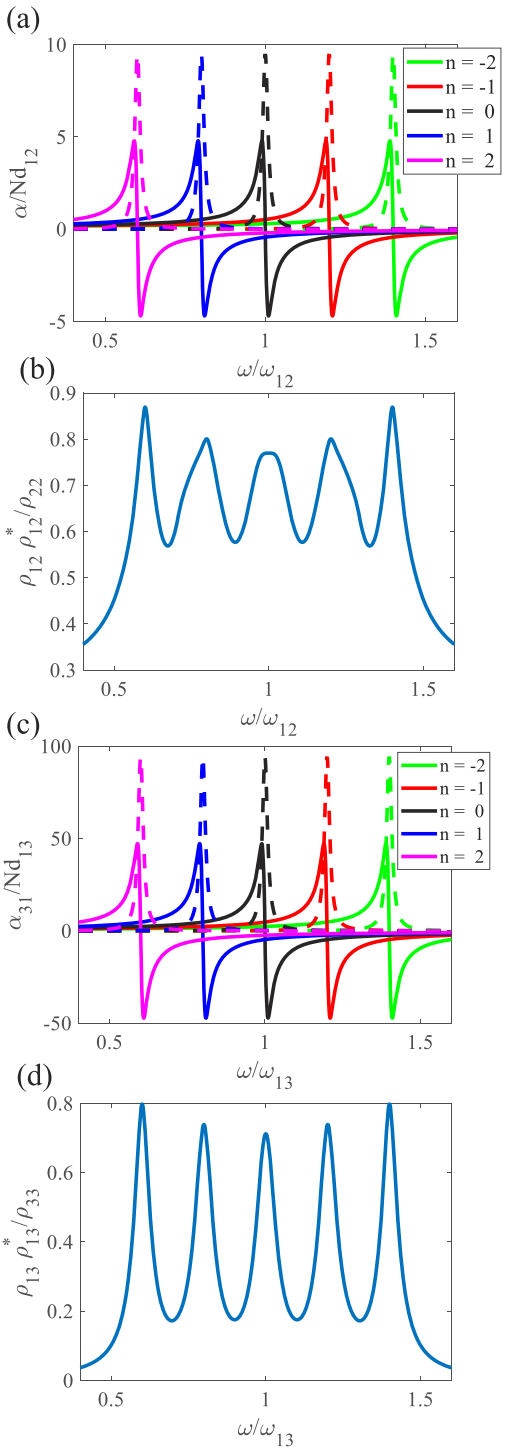


FIG. 3. Panels (a) and (c) show real (solid lines) and imaginary (dashed lines) α_n for two- and three-level systems, respectively, for $n = (-2, 2)$. The coherent and spontaneous emission ratio, given by $\rho_{12}\rho_{12}^*/\rho_{22}$ (b) and $\rho_{13}\rho_{13}^*/\rho_{33}$ (d) when all harmonics of the excitation have the same amplitude for two- and three-level systems.

dipoles oscillating at distinct frequencies, each corresponding to a specific harmonic. This occurs because polarizability is defined separately for each frequency harmonic, leading to a multi-frequency dipolar response. Moreover, each frequency harmonic corresponds to a different transition pathway in the three-level system, highlighting the role of time modulation in redistributing energy across multiple transitions and enhancing the overall control of the system. This effect allows for controlled energy exchange between different harmonics, leading to a tunable emission spectrum that can be engineered by adjusting the modulation frequency and amplitude.

IV. COUPLING QE WITH TM-QED

After calculating the polarizability of QEs, the next step to demonstrate the application of the proposed modeling is to place a 3 nm diameter V-shaped three-level quantum dot close to two time-modulated spheres with radii a and distanced $2d$ from each other, as shown in Fig. 4. In this example, the QE characteristics are chosen as $\lambda_{12} = 785$ nm, $\lambda_{13} = 850$ nm, $\Gamma_{12} = \Gamma_{13} = \Gamma_{23}/4 = 5 \times 10^{11}$ Hz, and $d_{12} = d_{13} = 10^{-30}$ cm. The quantum dot radius is embedded in a $d_{particle} = 2$ nm diameter silica sphere with a density of QE set as 10^{-23} cm^{-3}. Since the particle diameter is much smaller than the wavelength ($d_{particle}/\lambda_{13} = 0.0023$, well below the Rayleigh limit[59]), we considered a dipolar emission, as shown in Fig. 4. The time modulated sphere has an arbitrary permittivity $\varepsilon(t) = \varepsilon_r(1 + \delta \cos \omega_m t)$, where $\varepsilon_r = -4$, $\delta = 0.1$, and it is illuminated by a plane wave of 1 W/m^2 at λ_{12} with an electric field along the vector connecting the spheres center and operating at the weak coupling regime. It is important to notice that formalism can be adapted to various parameter choices, and the system is chosen to represent practical structures such as nano-antennas and metasurfaces close to QEs. To accommodate these scenarios, one needs to adjust the electromagnetic solver, which may involve using methods such as time-modulated DDA. Moreover, it is worth noting that our choice of material in this work is entirely arbitrary, allowing the focus on the proposed theoretical formalism, neglecting the effects of material properties, such as dispersion. One can obtain results for any specific material and toward a goal of interest, depending on the availability of materials. Nonetheless, various mechanisms such as acousto-optic[60,61] and free-carrier density modulation[62,63] can be employed to realize temporally modulated electromagnetic systems. In particular, indium tin oxide, titanium nitrate, and graphene have been used in combination with a plasmonic metasurface to demonstrate space–time modulation at optical and THz frequencies experimentally.[64–66]

Before exploring the time modulation (assuming $\delta = 0$), we use time-floquet Mie theory[42] to calculate the electric field at the quantum emitter (QE) position, providing the initial input for the iterative approach illustrated in Fig. 2. The Floquet–Mie framework used in this work builds upon a previously validated formalism, which was benchmarked against FDTD simulations and adiabatic limit comparisons, as detailed in Ref. 43. Using this framework combined with the model proposed in this manuscript, we calculated the dipolar emission at λ_{13} and the average emission $|\mathbf{p}_{13}(t)|$. Figures 5(a)–5(c) illustrate the Purcell factor (considering the emission), ρ_{33} and $\rho_{13}\rho_{13}^*/\rho_{33}$, for varying sphere radii and distances

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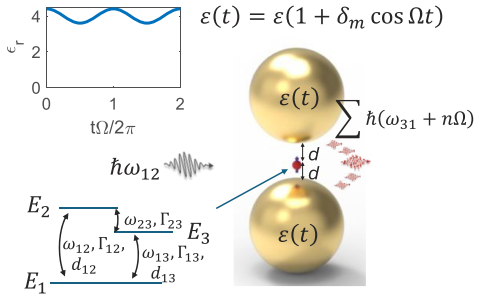


FIG. 4. Schematic of the proposed system. The 3 nm diameter V-shaped three-level quantum dot is placed between two time-modulated spheres with radius *a* and distanced 2*d*.

or RCWA. Moreover,

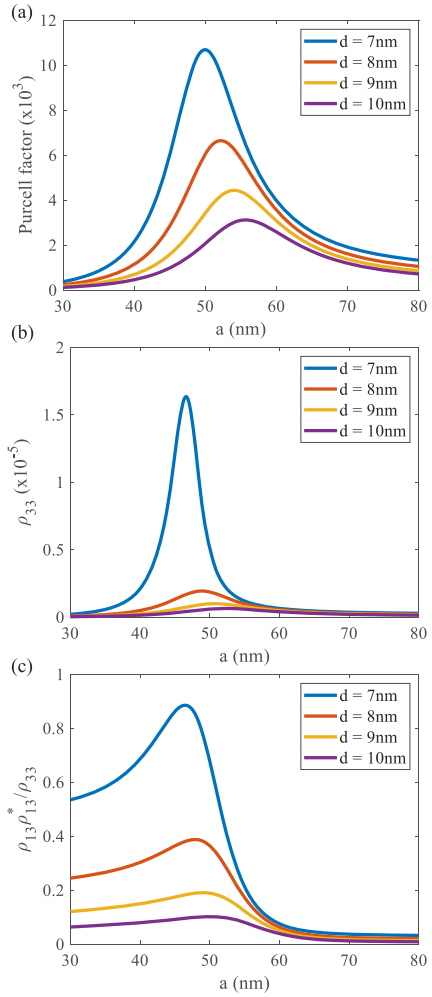


FIG. 5. Purcell factor (a) ρ_{33} (b) and $\rho_{13}\rho_{31}^*/\rho_{33}$ (c) for different sphere radius and distances from the QE.

between the QE and the spheres. As observed, closer proximity of the spheres to the QE enhances the Purcell factor due to intensified plasmonic resonances in the spheres. A stronger electric field increases the Rabi frequency, resulting in higher ρ_{33}. Notably, a strong resonance is achieved for a sphere radius a = 50 nm and d = 7 nm, as shown in Figs. 5(a) and 5(b), as observed by the high Purcell factor shown in Fig. 5(a). These parameters (a = 50 nm and d = 7 nm) optimize the system for enhanced spontaneous emission and provide a platform for controlling emission via time modulation. Note that the scattering properties of the metallic nanospheres used in the model were validated using Mie theory and FDTD simulations (see the supplementary material), confirming that they operate strictly within the electric dipole regime, which supports coupling through localized surface plasmon resonances that can be used to enhance the interaction of the structure with the QE. Moreover, the high Purcell indicates strong coupling between the quantum emitter and the surrounding structure, resulting in a more efficient energy transfer into the desired frequency harmonics. Moreover, in terms of emission, enhancing the Purcell factor is directly related to the QE lifetime,[7] resulting in faster modulation speeds that can be employed in telecommunications.

For the time-modulated scenario, we explore different modulation frequencies, ω_m = 4, 7, 10, and 14 Γ_{23}. Figure 6 shows the total power dissipated by the QE (P) normalized by the free-space power of the 23 transitions, $P_0 = d_{13}^2\omega_{13}^4/(4\pi\varepsilon_0 c^3)$ in Figs. 6(a)–6(e), the total radiated power (P_{rad}) in Figs. 6(f)–6(j), and[42] ρ_{33} in Figs. 6(k)–6(o) for the first two positive and negative frequency harmonics. The incident plane wave with frequency ω_{12} interacts with the time-modulated sphere system, generating harmonics that excite the QE. Moreover, the harmonics also contribute to exciting the QE, as can be seen by the harmonic excitation at Fig. 6(c), where for $n = \pm 1, \pm 2$, the harmonics generated by the structure at ω_{12} act to excite the QE.

In addition, the time-modulated Rabi frequency influences the electron population flux between levels 2 and 3, resulting in an oscillating population ρ_{33}, as shown by the harmonics in Figs. 6(k)–6(o). It is important to note that the population levels remain small due to the adoption of low-power excitation lasers. This choice has been made as the main focus of the work is on the development and validation of formalism, and since the system operates in the linear regime, increasing the excitation power would increase the population densities without altering the underlying dynamics. For completeness, we include simulations under higher-power excitation conditions and the population of ρ_{22} in the supplementary material.

The modulated population of ρ_{33} reflects the flux in which the electrons decay to level 1, as evidenced by the harmonics of the emission P in Figs. 6(a)–6(e). While most dissipated power is concentrated at the fundamental harmonic, the energy level oscillations cause the QE to emit photons at sideband harmonic frequencies.

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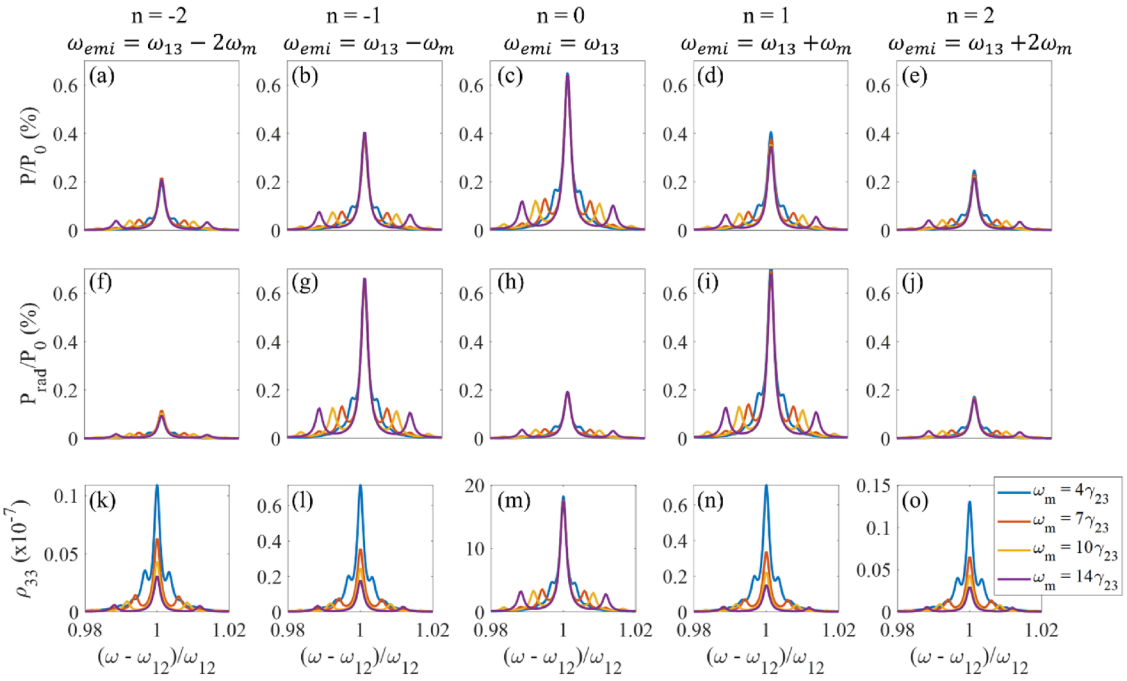


FIG. 6. Panels (a)–(e) show the normalized total dissipated power P/P_0, panels (f)–(j) show total radiated power P_{rad}/P_0 (k)–(n), and panels (k)–(o) show the excited state population ρ_{33}, for $n = 0, \pm1, \pm2$ and $\omega_m = 4\gamma_{23}, 7\gamma_{23}, 10\gamma_{23}$, and $14\gamma_{23}$.

Interestingly, the time-modulated system converts energy from the fundamental frequency to +1 and −1 harmonics, as evidenced by the higher radiated power at sidebands in Figs. 6(g) and (i) compared to the fundamental frequency shown in Fig. 6(h). Although the time modulation in this system is symmetric, optimization could enable power conversion to specific sidebands or frequencies. Notably, the quantum emitter (QE) excitation by a time-modulated field induces power dissipation across multiple frequency harmonics, which cannot be accurately captured when modeling the QE as a classical dipole. By incorporating time modulation into our quantum-mechanical framework, we introduce a new degree of freedom for controlling both the emission spectrum and radiative energy transfer of the QE, enabling dynamic spectral shaping through the proposed model.

In summary, we demonstrate that harmonic emission of quantum emitters (QEs) can be effectively excited and manipulated through their interaction with time-modulated (TM) systems. In addition, we show that the fundamental emission of the QE can be converted into specific sideband harmonics by carefully controlling the modulation of the sphere's permittivity. Unlike previous approaches that treat the emitter and environment separately, our framework inherently couples the QE dynamics with the time-modulated electromagnetic field, where the sidebands effectively act as additional energy levels, enabling new transition pathways and enhancing emission control. By exploring alternative modulation schemes, it is possible to enhance further and optimize the conversion efficiency of the radiated emission, offering a powerful tool for precise control over quantum light emission. Furthermore, by integrating this approach with arrays of nano-antennas, such as metasurfaces, not only can harmonic energy transfer be controlled but it also enables directional photon routing through beamforming. These results highlight the potential of time-modulated systems for advanced emission control.

It is important to emphasize that while previous *studies* have experimentally demonstrated time-modulated photonic structures, our work focuses predominantly on establishing a framework for coupling quantum emitter (QE) emission with dynamically modulated environments, which can be employed for any of these systems. The development and implementation of more practical time-modulated structures, such as plasmonic antennas or TMD, will be explored in future studies. In this context, promising material platforms include indium tin oxide (ITO), titanium nitride (TiN), doped silicon, and graphene, which have been experimentally shown to enable space–time modulation at optical and terahertz frequencies[64–66] and whose voltage-tunable permittivity allows for the realization of realistic and versatile designs, including at the permittivity regime presented here.

V. CONCLUSION

In conclusion, these results provide insights into the Markovian dynamics of open quantum systems interacting with time-modulated photonic structures, revealing how periodic modulation affects emission properties, coherence, and population transfer. We demonstrated that time modulation generates frequency harmonics that modify the system's energy levels, enabling the QE to radiate across multiple bands. By coupling the GKSL equation with the Floquet–Mie theory, we developed a robust approach to analyze the

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dipolar emission and Purcell factor in time-modulated systems. We also demonstrated that the emitted power of the quantum emitter (QE) can be decoupled from the total radiated power of the QE-TM system, highlighting how the time-modulated (TM) system enables energy transfer from the fundamental harmonic to higher-order harmonics. By optimizing the modulation parameters, it becomes possible to control the coupling to specific harmonics, effectively tuning the energy of the radiated photons. Furthermore, integrating this approach with arrays of nano-antennas, such as metasurfaces, enables active manipulation of emission properties post-fabrication, such as directional photon routing, opening new possibilities for tunable photonic devices and enhanced control over quantum and classical light sources. We provided a framework for designing time-modulated structures tailored for qubits, quantum transduction, and photonic systems. By utilizing the proposed system, it is possible to control the Purcell effect across different harmonics, allowing for modulation of emission lifetimes at multiple frequencies. The integration of tunable metasurfaces, time-modulated materials, and high-Q resonances provides a framework for advancements in photonic devices, with possible applications in quantum networks, secure communication, and high-speed data processing.

SUPPLEMENTARY MATERIAL

The supplementary material presents the full deduction for a two- and V-shaped 3-level system.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Achiles F. da Mota: Conceptualization (equal); Data curation (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Mohammad M. Sadafi**: Conceptualization (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Wei-Chi Chiu**: Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Bernardo Barbiellini**: Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Michael N. Leuenberger**: Investigation (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **Arun Bansil**: Funding acquisition (equal); Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Hossein Mosallaei**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

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